Basic Data Science

1. What is Data Science?

Data Science is the process of collecting, cleaning, analyzing, and interpreting data to extract useful insights using statistics, programming, and machine learning.

2. Difference between Data Science and Data Analytics?

- Data Science focuses on prediction, machine learning, and advanced modeling.
- Data Analytics mainly focuses on analyzing past data to find trends and insights.

3. What are the steps in a Data Science project lifecycle?

1. Data Collection
2. Data Cleaning
3. Exploratory Data Analysis (EDA)
4. Feature Engineering
5. Model Building
6. Model Evaluation
7. Deployment
8. Monitoring

4. What is supervised learning?

Supervised learning is a machine learning technique where the model learns from labeled data. Example: Predicting house prices.

5. What is unsupervised learning?

Unsupervised learning is a technique where the model works on unlabeled data to find patterns. Example: Customer segmentation.

6. Difference between classification and regression?

- Classification predicts categories.
- Regression predicts continuous values.

7. What is overfitting and underfitting?

- Overfitting: Model performs well on training data but poorly on test data.
- Underfitting: Model cannot capture patterns properly.

8. What is cross-validation?

Cross-validation is a technique used to evaluate model performance by dividing data into multiple parts.

9. What is feature engineering?

Feature engineering is the process of creating or modifying features to improve model performance.

10. What is feature selection?

Feature selection means selecting important features and removing unnecessary ones.

11. What are missing values and how do you handle them?

Missing values are empty data fields. Handling methods:

- Remove rows/columns
- Fill with mean, median, or mode
- Use prediction methods

12. What is data cleaning?

Data cleaning is the process of removing errors, duplicates, and inconsistencies from data.

13. Difference between structured and unstructured data?

- Structured data: Organized in rows and columns.
- Unstructured data: Images, videos, audio, text.

14. What is exploratory data analysis (EDA)?

EDA is the process of analyzing data using graphs and statistics to understand patterns and relationships.

15. What is normalization and standardization?

- Normalization scales data between 0 and 1.
- Standardization converts data into mean = 0 and standard deviation = 1.

16. What are outliers?

Outliers are data points that differ significantly from other observations.

17. What is bias and variance?

- Bias: Error due to simple assumptions.
- Variance: Error due to sensitivity to training data.

18. What is dimensionality reduction?

Dimensionality reduction reduces the number of input variables while keeping important information.

19. What is PCA?

PCA (Principal Component Analysis) is a dimensionality reduction technique that converts features into principal components.

20. What is a confusion matrix?

A confusion matrix is a table used to evaluate classification models.

21. Explain accuracy, precision, recall, and F1-score.

- Accuracy: Correct predictions/Total predictions
- Precision: Correct positive predictions/Total predicted positives
- Recall: Correct positive predictions/Actual positives
- F1-score: Harmonic mean of precision and recall

22. What is ROC-AUC?

ROC-AUC measures the performance of a classification model. Higher AUC means better model performance.

23. Difference between parametric and non-parametric models?

- Parametric models assume a fixed structure.
- Non-parametric models do not assume a fixed structure.

24. What is the curse of dimensionality?

As the number of features increases, model performance and efficiency decrease.

25. What is hypothesis testing?

Hypothesis testing is a statistical method used to make decisions based on data.

26. What is p-value?

P-value measures the probability that the observed result occurred by chance.

27. Difference between correlation and covariance?

- Correlation measures strength and direction.
- Covariance measures direction only.

28. What is probability distribution?

A probability distribution describes how values are distributed.

29. What is skewness?

Skewness measures the asymmetry of data distribution.

30. What is central limit theorem?

The central limit theorem states that the sampling distribution approaches normal distribution as sample size increases.

Machine Learning Questions

31. Explain Linear Regression.

Linear Regression predicts continuous values using a straight-line relationship.

32. Explain Logistic Regression.

Logistic Regression is used for binary classification problems.

33. Difference between Linear and Logistic Regression?

- Linear Regression predicts continuous values.
- Logistic Regression predicts categories.

34. What is Decision Tree?

Decision Tree is a supervised learning algorithm that splits data into branches for prediction.

35. What is Random Forest?

Random Forest is an ensemble method that combines multiple decision trees.

36. What is Naive Bayes?

Naive Bayes is a probabilistic algorithm based on Bayes theorem.

37. What is KNN algorithm?

KNN classifies data based on nearest neighbors.

38. What is SVM?

SVM (Support Vector Machine) is used for classification by finding the best separating hyperplane.

39. What is clustering?

Clustering groups similar data points together.

40. Explain K-Means clustering.

K-Means divides data into K clusters based on similarity.

41. What is ensemble learning?

Ensemble learning combines multiple models to improve performance.

42. What is bagging and boosting?

- Bagging reduces variance.
- Boosting reduces bias.

43. Difference between Random Forest and XGBoost?

- Random Forest builds trees independently.
- XGBoost builds trees sequentially.

44. What is gradient descent?

Gradient descent is an optimization algorithm used to minimize loss.

45. What is learning rate?

Learning rate controls how much the model updates during training.

46. What are hyperparameters?

Hyperparameters are settings defined before training the model.

47. What is regularization?

Regularization prevents overfitting by adding penalties.

48. Difference between L1 and L2 regularization?

- L1 removes unnecessary features.
- L2 reduces feature weights.

49. What is deep learning?

Deep learning is a subset of machine learning using neural networks with multiple layers.

50. What is neural network?

A neural network is a model inspired by the human brain.

Python Questions

51. Why is Python used in Data Science?

Python is easy to learn and has powerful libraries like NumPy, Pandas, and Scikit-learn.

52. What is NumPy?

NumPy is a Python library used for numerical computations.

53. What is Pandas?

Pandas is a Python library used for data manipulation and analysis.

54. Difference between loc and iloc in Pandas?

- loc uses labels.
- iloc uses index positions.

55. What is DataFrame?

A DataFrame is a two-dimensional table in Pandas.

56. How do you handle null values in Pandas?

Using:

- dropna()
- fillna()
- interpolation

57. What is Matplotlib?

Matplotlib is a Python library for data visualization.

58. What is Seaborn?

Seaborn is a visualization library built on Matplotlib.

59. Difference between list and NumPy array?

- Lists are slower.
- NumPy arrays are faster and memory efficient.

60. What is lambda function in Python?

A lambda function is an anonymous one-line function.

SQL Questions

61. What is SQL?

SQL is a language used to manage and query databases.

62. Difference between WHERE and HAVING?

- WHERE filters rows.
- HAVING filters grouped data.

63. What is JOIN?

JOIN combines rows from multiple tables.

64. Types of JOINs in SQL?

- INNER JOIN
- LEFT JOIN
- RIGHT JOIN
- FULL JOIN

65. Difference between DELETE, DROP, and TRUNCATE?

- DELETE removes rows.
- DROP removes entire table.
- TRUNCATE removes all rows quickly.

66. What is primary key and foreign key?

- Primary key uniquely identifies rows.
- Foreign key connects tables.

67. What is normalization in SQL?

Normalization organizes data to reduce redundancy.

68. What is GROUP BY?

GROUP BY groups rows with similar values.

69. What is subquery?

A subquery is a query inside another query.

70. Write a query to find the second highest salary.

```
SELECT MAX(salary)
FROM employees
WHERE salary < (SELECT MAX(salary) FROM employees);
```

Project-Based Questions

71. Explain your project briefly.

My project solves a real-world problem using machine learning and data analysis.

72. Why did you choose this project?

I chose this project to improve my practical skills and solve a useful problem.

73. What dataset did you use?

I used a public dataset collected from Kaggle.

74. How did you clean the data?

I removed duplicates, handled missing values, and normalized data.

75. Which algorithm did you use and why?

I used Random Forest because it gives good accuracy and handles overfitting well.

76. What challenges did you face?

Challenges included missing values and model tuning.

77. How did you evaluate your model?

Using accuracy, precision, recall, and F1-score.

78. What accuracy did you get?

I achieved around 85–95% accuracy depending on the dataset.

79. How can your project be improved?

By using larger datasets and better feature engineering.

80. How would you deploy your model?

Using Flask/FastAPI with cloud platforms like Render or AWS.

HR and Scenario Questions

81. Tell me about yourself.

I am a B.Tech IT student passionate about Data Science, Machine Learning, and Backend Development.

82. Why do you want to become a Data Scientist?

I enjoy solving problems using data and building intelligent systems.

83. What are your strengths and weaknesses?

- Strengths: Quick learner, problem-solving.
- Weakness: Sometimes spend too much time perfecting details.

84. Describe a challenging problem you solved.

I solved issues related to missing data and improved model accuracy through preprocessing.

85. How do you handle tight deadlines?

I prioritize tasks and maintain proper time management.

86. Have you worked in a team project?

Yes, I have collaborated with teammates during academic projects and hackathons.

87. Why should we hire you?

I have strong technical skills, learning ability, and dedication.

88. Where do you see yourself in 5 years?

I see myself as an experienced Data Scientist working on impactful projects.

89. How do you stay updated with new technologies?

I learn through online courses, documentation, YouTube, and projects.

90. Explain a project to a non-technical person.

I built a system that analyzes data and predicts outcomes to help users make better decisions.